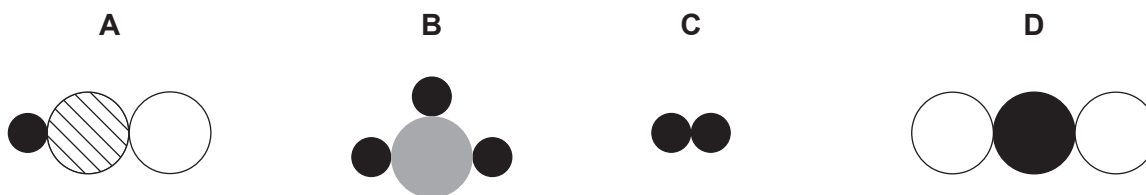


2. Diagrams **A**, **B**, **C** and **D** represent hydrogen (H_2), sulfur dioxide (SO_2), hypochlorous acid (HClO) and phosphine (PH_3) but **not in that order**.



- (a) (i) Give the **letter** of the diagram that represents an element. Give a reason for your answer. [2]

Letter

Reason

- (ii) Give the **letter** of the diagram that represents SO_2 . [1]

.....

- (iii) Use diagrams **A**, **B**, **C** and **D** above to work out which of the following diagrams represents water (H_2O). **Circle** the correct diagram. [1]



- (b) (i) Calculate the relative molecular mass (M_r) of hypochlorous acid, HClO. [1]

$$A_r(\text{H}) = 1$$

$$A_r(\text{O}) = 16$$

$$A_r(\text{Cl}) = 35.5$$

$$M_r = \dots\dots\dots$$

- (ii) The relative molecular mass (M_r) of sulfur dioxide is 64.

Calculate the percentage by mass of sulfur in sulfur dioxide, SO_2 . [2]

$$A_r(\text{S}) = 32$$

$$\text{Percentage} = \dots\dots\dots \%$$

- (c) Phosphine (PH_3) is made when phosphorus and hydrogen react.

Put a tick (✓) in the box next to the correctly balanced symbol equation for this reaction. [1]

